

Technical Analysis System

Functional Specification

System: Portfolio Tracker – Technical Analysis Module

Version: 2.0

Date: October 2025

Status: Updated Specification

Document Control

Version	Date	Changes	Author
1.0	October 2025	Initial functional specification	Original Team
2.0	October 2025	Major revision: Added volume analysis, rebalanced weights, improved scoring logic, added confidence metrics	Functional Analysis Team

1. Executive Summary

The Technical Analysis System provides automated stock analysis using multiple technical indicators to generate buy/sell/hold signals with confidence metrics. Version 2.0 introduces significant improvements including volume analysis, enhanced scoring logic, and transparent confidence reporting.

Key Improvements in Version 2.0

- Volume analysis integration for signal confirmation
- Stochastic Oscillator for enhanced momentum detection
- Full Bollinger Bands utilization with defined scoring rules
- Rebalanced indicator weights (total: 30 points)
- Refined RSI interpretation separating momentum from reversal signals
- Confidence scoring based on data availability
- Detailed signal breakdown for transparency
- More granular signal classifications

2. System Overview

2.1 Purpose

The system helps investors make informed decisions by analyzing stock price movements and volume through mathematical indicators that identify trends, momentum, volatility, and potential reversal points. The system provides not only a signal but also the reasoning and confidence level behind that signal.

2.2 Key Features

- Multi-indicator technical analysis (9 indicators)

- Volume-confirmed signals
- Weighted scoring algorithm with rebalanced weights
- Seven-level signal classification
- Confidence metrics based on data availability and indicator agreement
- Transparent signal breakdown showing contributing indicators
- Automatic calculation from historical price and volume data
- Flexible data requirements with graceful degradation
- Divergence detection capabilities

2.3 Design Principles

- **Transparency:** Users see which indicators contributed to the signal
- **Confidence:** System reports reliability based on data completeness and indicator agreement
- **Context:** Signals consider market regime and trend strength
- **Balance:** No single indicator dominates; confirmation required from multiple sources
- **Practicality:** Adapts to available data while maintaining quality standards

3. Technical Indicators

The system calculates nine primary technical indicators, each serving a specific purpose in market analysis. Indicators are grouped into categories: Trend, Momentum, Volatility, and Volume.

3.1 Trend Indicators

3.1.1 Simple Moving Average (SMA)

Purpose: Identifies price trends by smoothing out short-term fluctuations.

Calculation Periods:

- **SMA 20:** 20-day moving average (short-term trend)
- **SMA 50:** 50-day moving average (medium-term trend)
- **SMA 200:** 200-day moving average (long-term trend)

Interpretation:

- Price above SMA → Bullish signal (upward trend)
- Price below SMA → Bearish signal (downward trend)

Data Requirements:

- SMA 20: Minimum 25 price points (20 for calculation + 5 warm-up)
- SMA 50: Minimum 55 price points (50 for calculation + 5 warm-up)
- SMA 200: Minimum 205 price points (200 for calculation + 5 warm-up)

3.1.2 Exponential Moving Average (EMA)

Purpose: Similar to SMA but gives more weight to recent prices, making it more responsive to new information.

Calculation Periods:

- **EMA 12:** 12-day exponential moving average
- **EMA 26:** 26-day exponential moving average

Interpretation:

- Used primarily for MACD calculation
- More sensitive to recent price changes than SMA

Data Requirements:

- EMA 12: Minimum 15 price points
- EMA 26: Minimum 30 price points

3.1.3 Golden Cross / Death Cross

Purpose: Identifies major trend reversals based on moving average crossovers.

Patterns:

- **Golden Cross:** SMA 50 crosses above SMA 200 → Strong bullish signal
- **Death Cross:** SMA 50 crosses below SMA 200 → Strong bearish signal

Interpretation:

- SMA 50 > SMA 200 → Long-term bullish trend
- SMA 50 < SMA 200 → Long-term bearish trend

Data Requirements:

- Both SMA 50 and SMA 200 must be available (minimum 205 price points)

3.2 Momentum Indicators UPDATED

3.2.1 Relative Strength Index (RSI) UPDATED

Purpose: Measures the speed and magnitude of price changes to identify overbought or oversold conditions and momentum strength.

Calculation Period: 14 days (RSI 14)

Scale: 0 to 100

Version 2.0 Change: Enhanced RSI Interpretation

RSI signals are now categorized into momentum signals and reversal signals to provide more accurate context.

Interpretation:

RSI Range	Signal Type	Meaning	Points
< 30	Reversal Signal	Oversold - potential bounce opportunity	+4 bullish
30-50	Momentum Signal	Below neutral - bearish momentum	+2 bearish
50-70	Momentum Signal	Above neutral - bullish momentum	+2 bullish
> 70	Reversal Signal	Overbought - potential correction	+4 bearish

Data Requirements:

- Minimum 20 price points (14 for calculation + 6 warm-up period)

3.2.2 Moving Average Convergence Divergence (MACD) UPDATED

Purpose: Shows the relationship between two exponential moving averages to identify trend changes and momentum.

Components:

- **MACD Line:** Difference between 12-day EMA and 26-day EMA
- **Signal Line:** 9-day EMA of the MACD line

Version 2.0 Change: Removed MACD Histogram

The histogram has been removed to eliminate redundancy since it represents the same information as the MACD/Signal relationship.

Interpretation:

- MACD > Signal: Bullish momentum
- MACD < Signal: Bearish momentum

Data Requirements:

- Minimum 40 price points (26 + 9 + 5 warm-up)

3.2.3 Stochastic Oscillator NEW

Purpose: Compares a stock's closing price to its price range over a specific period to identify momentum and potential reversal points.

Calculation:

- **%K Line:** 14-period stochastic
- **%D Line:** 3-period SMA of %K (signal line)

Scale: 0 to 100

Interpretation:

Condition	Signal	Points
%K < 20	Oversold - bullish reversal potential	+3 bullish
%K > 80	Overbought - bearish reversal potential	+3 bearish
%K crosses above %D (20-80 range)	Bullish momentum	+1.5 bullish
%K crosses below %D (20-80 range)	Bearish momentum	+1.5 bearish

Data Requirements:

- Minimum 20 price points (14 + 3 + 3 warm-up)

3.3 Volatility Indicators

UPDATED

3.3.1 Bollinger Bands

UPDATED

Purpose: Measures market volatility and identifies potential price extremes and reversal points.

Components:

- **Middle Band:** 20-day SMA
- **Upper Band:** Middle + (2 × standard deviation)
- **Lower Band:** Middle - (2 × standard deviation)

Version 2.0 Enhancement

Bollinger Bands now have explicit scoring rules and are fully integrated into the signal generation algorithm.

Interpretation & Scoring:

Price Position	Signal	Points
Price > Upper Band	Potentially overbought - caution	+3 bearish
Price < Lower Band	Potentially oversold - opportunity	+3 bullish
Price between Middle and Upper	Bullish zone	+1.5 bullish
Price between Middle and Lower	Bearish zone	+1.5 bearish
Bandwidth expanding	Increasing volatility - trend confirmation	Context dependent
Bandwidth contracting (squeeze)	Decreasing volatility - breakout pending	Neutral (awaiting direction)

Data Requirements:

- Minimum 25 price points (20 for SMA + 5 warm-up)

3.4 Volume Indicators

NEW

3.4.1 Volume Trend Analysis

NEW

Purpose: Confirms price movements by analyzing trading volume. Volume validates the strength of price trends.

Calculation:

- Compare current volume to 20-day average volume
- Analyze volume direction with price direction

Why Volume Matters

Price increases on high volume are more reliable than those on low volume. Volume confirms the conviction behind price movements.

Interpretation & Scoring:

Condition	Signal	Points
Price ↑ + Volume ↑ (>20% above avg)	Strong bullish confirmation	+4 bullish
Price ↓ + Volume ↑ (>20% above avg)	Strong bearish confirmation	+4 bearish
Price ↑ + Volume ↓ (<20% below avg)	Weak bullish move - caution	+1 bearish
Price ↓ + Volume ↓ (<20% below avg)	Weak bearish move - potential reversal	+1 bullish
Volume within normal range (±20%)	Neutral - no strong conviction	0 points

Data Requirements:

- Minimum 25 volume data points (20 for average + 5 for trend)

4. Signal Generation Algorithm UPDATED

4.1 Weighted Scoring System UPDATED

The system uses a rebalanced weighted points system where each indicator contributes points based on its reliability, importance, and to avoid redundancy.

Major Change from Version 1.0

Total weight increased from 23 to 30 points for cleaner mathematics. MACD histogram removed, volume and stochastic added, RSI and Bollinger weights adjusted.

Weight Distribution:

Indicator	Weight	Change	Rationale
Golden/Death Cross	5	+1	Strongest long-term trend signal
Volume Trend	4	NEW	Essential confirmation of price moves
RSI 14	4	-1	High reliability but not dominant
SMA 20 vs Price	3	0	Short-term trend indicator
SMA 50 vs Price	3	0	Medium-term trend indicator
MACD Signal	3	-1	Reliable momentum, reduced to avoid double-counting
Stochastic Oscillator	3	NEW	Momentum and reversal confirmation
Total Maximum Weight	30	+7	Cleaner calculation basis

Indicator	Weight	Change	Rationale
Bollinger Bands Position	3	NEW	Volatility context and extremes
SMA 200 vs Price	2	0	Long-term trend baseline
MACD Histogram	-	REMOVED	Redundant with MACD signal
Total Maximum Weight	30	+7	Cleaner calculation basis

4.2 Point Allocation Rules UPDATED

Trend Indicators (Moving Averages):

- **SMA 20 vs Price:**
Current Price > SMA 20: +3 bullish points
Current Price < SMA 20: +3 bearish points
- **SMA 50 vs Price:**
Current Price > SMA 50: +3 bullish points
Current Price < SMA 50: +3 bearish points
- **SMA 200 vs Price:**
Current Price > SMA 200: +2 bullish points
Current Price < SMA 200: +2 bearish points
- **Golden/Death Cross:**
SMA 50 > SMA 200: +5 bullish points
SMA 50 < SMA 200: +5 bearish points

Momentum Indicators:

- **RSI Analysis** UPDATED
RSI < 30 (Oversold): +4 bullish points
RSI > 70 (Overbought): +4 bearish points
30 ≤ RSI ≤ 70 and RSI > 50: +2 bullish points
30 ≤ RSI ≤ 70 and RSI < 50: +2 bearish points
- **MACD Analysis** UPDATED
MACD > Signal Line: +3 bullish points
MACD < Signal Line: +3 bearish points
- **Stochastic Oscillator** NEW
%K < 20: +3 bullish points
%K > 80: +3 bearish points
%K crosses above %D (mid-range): +1.5 bullish points
%K crosses below %D (mid-range): +1.5 bearish points

Volatility Indicators:

- **Bollinger Bands** UPDATED
Price > Upper Band: +3 bearish points
Price < Lower Band: +3 bullish points
Price between Middle and Upper: +1.5 bullish points
Price between Middle and Lower: +1.5 bearish points

Volume Indicators:

- **Volume Trend** NEW
Price ↑ + Volume ↑ (>20% above avg): +4 bullish points
Price ↓ + Volume ↑ (>20% above avg): +4 bearish points
Price ↑ + Volume ↓ (<20% below avg): +1 bearish points
Price ↓ + Volume ↓ (<20% below avg): +1 bullish points

4.3 Score Calculation UPDATED

Base Net Score Calculation:

Base Score = (Bullish Points - Bearish Points) / Total Available Weight

Range: -1.0 to +1.0

Confidence Modifier NEW :

Confidence Factor = Number of Available Indicators / Total Possible Indicators (9)

Final Adjusted Score:

Final Score = Base Score × (0.7 + 0.3 × Confidence Factor)

Why Confidence Adjustment?

Scores based on limited data are less reliable. The confidence modifier ensures that:

- With all 9 indicators (100% confidence): Full score applies (multiplier = 1.0)
- With 5 indicators (55% confidence): Score × 0.87
- With 3 indicators (33% confidence): Score × 0.80

This prevents overconfident signals from incomplete data.

Example Calculation:

Scenario:

- Bullish Points: 18
- Bearish Points: 5
- Available Weight: 25 (7 out of 9 indicators available)
- Confidence Factor: 7/9 = 0.778

Calculations:

Base Score = (18 - 5) / 25 = 0.52
Confidence Modifier = 0.7 + (0.3 × 0.778) = 0.933
Final Score = 0.52 × 0.933 = 0.485

Result: BUY signal (score 0.485 falls in 0.3-0.6 range)

4.4 Signal Classification UPDATED

Version 2.0 Enhancement

Expanded from 5 to 7 signal levels for more granular guidance

Net Score Range	Signal	Meaning	Recommended Action
≥ 0.6	STRONG BUY	Very bullish - multiple strong positive indicators	Consider buying or adding to position
0.3 to 0.6	BUY	Moderately bullish - positive trend indicated	Consider buying with normal position sizing
0.1 to 0.3	WEAK BUY	Slightly bullish - early positive signals	Monitor closely, small position if desired
-0.1 to 0.1	HOLD	Neutral - no clear trend or mixed signals	Maintain current position, no action needed

Net Score Range	Signal	Meaning	Recommended Action
-0.3 to -0.1	WEAK SELL	Slightly bearish - early negative signals	Consider reducing position or taking profits
-0.6 to -0.3	SELL	Moderately bearish - negative trend indicated	Consider selling or reducing position
< -0.6	STRONG SELL	Very bearish - multiple strong negative indicators	Consider exiting position

4.5 Signal Confidence Rating NEW

In addition to the signal itself, the system provides a confidence rating based on data completeness and indicator agreement.

Confidence Level	Criteria	Display
HIGH	8-9 indicators available AND ≥75% agreement	☆☆☆☆
MEDIUM	6-7 indicators available OR 60-74% agreement	☆☆☆
LOW	<6 indicators available OR <60% agreement	☆☆

Agreement Calculation:

Agreement % = (Dominant Direction Points / Total Points) × 100

Where dominant direction is the side (bullish or bearish) with more points

5. Adaptive Data Requirements UPDATED

5.1 Flexibility

The system adapts to available data, calculating only indicators for which sufficient data exists. This ensures the system remains functional across different data availability scenarios.

5.2 Minimum Data Requirements

Indicator	Minimum Price Points	Volume Required
SMA 20, Bollinger Bands	25	No
RSI 14, Stochastic	20	No
MACD	40	No
SMA 50	55	No
SMA 200, Golden/Death Cross	205	No
Volume Trend	25	Yes (25 points)

5.3 Operational Tiers

Tier 1: Minimal (20-24 price points)

Available Indicators: RSI, Stochastic (2/9 indicators)

Max Weight: 7 points

Confidence: LOW ★

System will generate signal but flag as low confidence

Tier 2: Basic (25-39 price points + volume)

Available Indicators: SMA 20, RSI, Stochastic, Bollinger Bands, Volume (5/9 indicators)

Max Weight: 17 points

Confidence: LOW to MEDIUM ★ - ★★

Sufficient for short-term analysis

Tier 3: Standard (40-54 price points + volume)

Available Indicators: All except SMA 50, 200, and Golden/Death Cross (6/9 indicators)

Max Weight: 22 points

Confidence: MEDIUM ★★

Good for medium-term analysis

Tier 4: Enhanced (55-204 price points + volume)

Available Indicators: All except SMA 200 and Golden/Death Cross (7/9 indicators)

Max Weight: 25 points

Confidence: MEDIUM to HIGH ★★★ - ★★★★★

Excellent for most analysis needs

Tier 5: Complete (205+ price points + volume)

Available Indicators: All 9 indicators

Max Weight: 30 points

Confidence: HIGH ★★★★★

Optimal - full long-term analysis capability

5.4 Graceful Degradation

When insufficient data exists for certain indicators:

- 1. Those indicators are excluded from the calculation
- 2. Total available weight is adjusted accordingly
- 3. Signal is generated from available indicators
- 4. Confidence rating reflects data completeness
- 5. Output clearly indicates which indicators were used/missing
- 6. If fewer than 2 indicators available, system returns "INSUFFICIENT DATA" instead of HOLD

6. Output Specifications NEW SECTION

6.1 Primary Output Structure

The system must provide transparent, actionable output that shows not just the signal, but the reasoning behind it.

Required Output Fields:

Field	Type	Description
signal	String	STRONG_BUY BUY WEAK_BUY HOLD WEAK_SELL SELL STRONG_SELL
score	Number	Final adjusted score (-1.0 to +1.0)
baseScore	Number	Base score before confidence adjustment
confidence	String	HIGH MEDIUM LOW

Field	Type	Description
confidenceStars	Number	1-3 stars
bullishPoints	Number	Total bullish points accumulated
bearishPoints	Number	Total bearish points accumulated
availableWeight	Number	Total weight of available indicators
indicatorsUsed	Number	Count of indicators with sufficient data (out of 9)
agreement	Number	Percentage of indicators agreeing with dominant direction
breakdown	Object	Detailed indicator-by-indicator results
warnings	Array	Any cautions or notes about the analysis

6.2 Breakdown Structure

The breakdown object provides transparency by showing each indicator's contribution:

```
breakdown: { trend: { sma20: { available: true, signal: "bullish", points: 3, value: 145.2 }, sma50: { available: true, signal: "bullish", points: 3, value: 140.8 }, sma200: { available: true, signal: "bullish", points: 2, value: 135.1 }, goldenCross: { available: true, signal: "bullish", points: 5, active: true } }, momentum: { rsi: { available: true, signal: "bullish", points: 2, value: 62.3, category: "momentum" }, macd: { available: true, signal: "bullish", points: 3, macdValue: 2.5, signalValue: 1.8 }, stochastic: { available: true, signal: "bullish", points: 1.5, kValue: 65, dValue: 60 } }, volatility: { bollinger: { available: true, signal: "bullish", points: 1.5, position: "upper_half", upper: 155, middle: 145, lower: 135 } }, volume: { volumeTrend: { available: true, signal: "bullish", points: 4, currentVol: 2500000, avgVol: 2000000, change: "+25%" } } }
```

6.3 Example Complete Output

Stock: XYZ

Signal: BUY ★★★★★

Score: +0.48 | Confidence: HIGH | Agreement: 82%

Signal Breakdown (7/9 indicators available):

Bullish Signals (22 points):

- ✓ Golden Cross active (+5)
- ✓ Volume increasing with price (+4)
- ✓ Price above SMA 20, 50, 200 (+8)
- ✓ MACD bullish crossover (+3)
- ✓ RSI showing bullish momentum (+2)

Bearish Signals (4 points):

- ✗ Stochastic entering overbought (+3 bearish)
- ✗ Price approaching upper Bollinger Band (+1 caution)

⚠ **Considerations:**

- Stochastic and Bollinger suggest potential short-term pullback
- Strong uptrend intact; consider waiting for dip to add position

Missing: SMA 200 and Bollinger Bands (insufficient historical data)

6.4 Warning Messages

The system should generate contextual warnings when appropriate:

- **Low confidence:** "Signal based on limited data (only X/9 indicators available)"
- **Mixed signals:** "Indicators show divergence - trend signals bullish but momentum weakening"
- **Overbought in uptrend:** "Strong uptrend but overbought conditions suggest potential pullback"
- **Low volume:** "Price movement not confirmed by volume - exercise caution"
- **Insufficient data:** "Less than 20 days of data - unable to generate reliable signal"
- **High volatility:** "Bollinger Bands expanding - increased volatility and risk"

7. Example Scenarios UPDATED

7.1 Complete Data - Strong Bullish Signal

Input Data:

- 250 days of price and volume history for Stock XYZ
- Current Price: \$150
- Price trend: Strong upward over past months
- Volume: Increasing with price rises

Calculated Indicators:

Indicator	Value	Signal	Points
SMA 20	\$145	Bullish (price above)	+3
SMA 50	\$140	Bullish (price above)	+3
SMA 200	\$135	Bullish (price above)	+2
Golden Cross	Active	Bullish	+5
RSI 14	65	Bullish momentum	+2
MACD	2.5 vs 1.8	Bullish	+3
Stochastic	%K:70 %D:65	Bullish (K>D)	+1.5
Bollinger	Upper half	Bullish zone	+1.5
Volume	+30% vs avg	Strong bullish confirmation	+4

Calculation:

- Total Bullish: 25 points
- Total Bearish: 0 points
- Available Weight: 30 points (9/9 indicators)
- Base Score: $(25 - 0) / 30 = 0.833$
- Confidence Factor: $9/9 = 1.0$
- Final Score: $0.833 \times (0.7 + 0.3 \times 1.0) = 0.833$
- Agreement: 100% (all bullish)

Result: STRONG BUY ★★★★★

High confidence - all indicators confirm strong uptrend with volume support

7.2 Limited Data - Moderate Signal

Input Data:

- 45 days of price and volume history for Stock ABC
- Current Price: \$82
- Recent upward momentum

Available Indicators (6/9):

Indicator	Value	Signal	Points
SMA 20	\$80	Bullish	+3
RSI 14	58	Bullish momentum	+2
MACD	1.2 vs 1.0	Bullish	+3
Stochastic	%K:60 %D:55	Bullish	+1.5
Bollinger	Middle-Upper	Bullish	+1.5

Indicator	Value	Signal	Points
Volume	+15% vs avg	Neutral	0
SMA 50	-	N/A	-
SMA 200	-	N/A	-
Golden Cross	-	N/A	-

Calculation:

- Total Bullish: 11 points
- Total Bearish: 0 points
- Available Weight: 17 points (6/9 indicators)
- Base Score: $(11 - 0) / 17 = 0.647$
- Confidence Factor: $6/9 = 0.667$
- Final Score: $0.647 \times (0.7 + 0.3 \times 0.667) = 0.583$
- Agreement: 100% (all bullish, but limited indicators)

Result: BUY ⭐⭐

Medium confidence - positive signals but missing long-term trend confirmation

⚠ Warning: Long-term trend indicators unavailable due to limited history

7.3 Mixed Signals - Hold Scenario

Input Data:

- 210 days of price and volume history for Stock DEF
- Current Price: \$95
- Price consolidating, mixed indicators

Calculated Indicators:

Indicator	Value	Signal	Points
SMA 20	\$96	Bearish (price below)	+3 bearish
SMA 50	\$94	Bullish (price above)	+3 bullish
SMA 200	\$92	Bullish (price above)	+2 bullish
Golden Cross	Active	Bullish	+5 bullish
RSI 14	48	Bearish momentum	+2 bearish
MACD	0.8 vs 1.1	Bearish	+3 bearish
Stochastic	%K:45 %D:50	Bearish (K<D)	+1.5 bearish
Bollinger	Lower half	Bearish zone	+1.5 bearish
Volume	-5% vs avg	Neutral	0

Calculation:

- Total Bullish: 10 points
- Total Bearish: 11 points
- Available Weight: 30 points (9/9 indicators)
- Base Score: $(10 - 11) / 30 = -0.033$
- Confidence Factor: $9/9 = 1.0$
- Final Score: $-0.033 \times (0.7 + 0.3 \times 1.0) = -0.033$
- Agreement: 52% (nearly split)

Result: HOLD ⭐⭐⭐

High confidence in *HOLD* - trend is bullish but momentum turning bearish
⚠ Warning: Mixed signals - long-term trend bullish but short-term momentum weakening. Monitor closely.

7.4 Bearish with Volume Confirmation

Input Data:

- 250 days of price and volume history for Stock GHI
- Current Price: \$45
- Declining price with increasing volume

Calculated Indicators:

Indicator	Value	Signal	Points
SMA 20	\$48	Bearish	+3 bearish
SMA 50	\$51	Bearish	+3 bearish
SMA 200	\$55	Bearish	+2 bearish
Golden Cross	Death Cross	Bearish	+5 bearish
RSI 14	35	Bearish momentum	+2 bearish
MACD	-1.5 vs -0.8	Bearish	+3 bearish
Stochastic	%K:25 %D:30	Neutral/Oversold	+3 bullish
Bollinger	Below lower band	Oversold signal	+3 bullish
Volume	+35% on decline	Strong bearish confirmation	+4 bearish

Calculation:

- Total Bullish: 6 points (oversold signals)
- Total Bearish: 22 points
- Available Weight: 30 points (9/9 indicators)
- Base Score: $(6 - 22) / 30 = -0.533$
- Confidence Factor: $9/9 = 1.0$
- Final Score: $-0.533 \times (0.7 + 0.3 \times 1.0) = -0.533$
- Agreement: 73% bearish

Result: SELL ★★☆☆

High confidence - strong downtrend with volume confirmation
⚠ Note: Oversold signals present but downtrend remains strong. Not yet a reversal.

8. Use Cases

8.1 Portfolio Review

Scenario: User wants to review all holdings for potential rebalancing

System Behavior:

- Calculate technical signals for each position in portfolio
- Display signals with confidence ratings
- Identify positions with SELL or STRONG_SELL signals
- Flag positions with low confidence due to limited data
- Sort by signal strength or by positions requiring attention
- Provide summary: "3 positions showing SELL signals, 5 showing BUY"

8.2 New Investment Research

Scenario: User researching potential new stock purchase

System Behavior:

- Analyze technical indicators for candidate stock
- Provide detailed signal breakdown showing which indicators support the signal
- Display confidence rating based on data availability
- Highlight any warnings (overbought, low volume, mixed signals)
- Show trend context (short vs. long-term alignment)
- Suggest: "Strong uptrend but consider entry on pullback" or "Good entry point - oversold with trend support"

8.3 Exit Strategy Monitoring

Scenario: User holds profitable position and wants to monitor for exit signals

System Behavior:

- Track signal changes over time
- Alert when signal changes from BUY to WEAK_BUY or HOLD
- Flag early warning signs (momentum divergence, volume declining)
- Notify on SELL or STRONG_SELL signals
- Provide context: "Uptrend intact but momentum weakening - monitor closely"

8.4 Watchlist Screening

Scenario: User monitors watchlist of 20+ stocks for opportunities

System Behavior:

- Batch calculate signals for all watchlist stocks
- Filter and sort by signal strength
- Highlight new STRONG_BUY or BUY signals
- Show confidence ratings to prioritize research
- Provide quick summary view with key metrics per stock

9. Limitations and Considerations

9.1 Data Quality Dependencies

- **Accuracy depends on input:** System requires clean, accurate historical price and volume data
- **Historical data only:** Cannot predict unexpected events (earnings surprises, regulatory changes, macroeconomic shocks)
- **Market hours limitation:** Only reflects traded prices, not after-hours movements
- **Corporate actions:** Stock splits, dividends must be adjusted in source data

9.2 Not Financial Advice

Important Disclaimer

- **Tool for analysis only:** Not a recommendation to buy or sell securities
- **One perspective:** Should be combined with fundamental analysis, valuation, company research
- **User responsibility:** Final investment decisions rest with the user
- **No guarantees:** Past price patterns do not guarantee future results
- **Risk awareness:** All investing involves risk of loss

9.3 Market Condition Sensitivities

- **Trending markets:** System works best in clear trending markets (up or down)
- **Sideways markets:** Less reliable in ranging/choppy markets with no clear trend
- **Lagging indicators:** Based on past prices, not predictive of future moves
- **External factors:** Cannot account for news, earnings, analyst reports, sector rotation, market sentiment
- **Low liquidity:** Less reliable for thinly traded stocks where price gaps are common

9.4 Signal Interpretation Guidelines

- **Confidence matters:** Low confidence signals should be treated with caution
- **Context is key:** Consider overall market conditions and sector trends
- **Timeframe alignment:** Day traders and long-term investors may interpret same signal differently
- **Multiple timeframes:** Consider analyzing multiple timeframes (daily, weekly) for confirmation
- **Risk management:** Use position sizing and stop losses regardless of signal strength

9.5 Known Edge Cases

- **Recent IPOs:** Limited history may produce unreliable signals
- **Penny stocks:** High volatility and low liquidity reduce reliability
- **Market crashes:** Extreme volatility can produce whipsaw signals
- **Gap events:** Overnight gaps (earnings, takeover bids) not captured by indicators
- **Manipulation:** System cannot detect price manipulation or unusual activity

10. System Benefits

10.1 Objectivity and Consistency

- Removes emotional bias from technical analysis
- Applies consistent rules across all analyzed securities
- Quantifiable decision criteria enable backtesting
- Documented methodology ensures reproducible results

10.2 Efficiency and Scalability

- Instant analysis of any stock with available data
- Batch processing enables portfolio-wide analysis
- Automated recalculation with new daily data
- Scales from single stock to hundreds of securities

10.3 Transparency and Learning ENHANCED

- Clear breakdown shows which indicators contributed to signal
- Users understand the "why" behind each signal
- Educational value helps users learn technical analysis concepts
- Confidence metrics build appropriate trust levels
- Warnings highlight important considerations

10.4 Risk Management Support

- Early warning system for position exits
- Identifies weakening trends before major reversals
- Confidence ratings help with position sizing decisions
- Multiple confirmation reduces false signals

11. Future Enhancements

11.1 Phase 2: Advanced Indicators

- **On-Balance Volume (OBV):** Cumulative volume indicator for trend confirmation
- **Volume-Weighted Average Price (VWAP):** Intraday benchmark for institutional trading levels
- **Average True Range (ATR):** Volatility measurement for stop loss placement
- **Fibonacci Retracement:** Key support/resistance levels based on mathematical ratios
- **Ichimoku Cloud:** Comprehensive trend and momentum system

11.2 Phase 3: Pattern Recognition

- Chart pattern detection (head & shoulders, double tops/bottoms, triangles, flags)
- Candlestick pattern analysis (doji, hammer, engulfing patterns)
- Support and resistance level identification
- Trendline detection and breakout alerts
- Price channel analysis

11.3 Phase 4: Customization and Profiles

- **User-adjustable weights:** Allow users to customize indicator importance
- **Trading style profiles:** Pre-configured settings for day trading, swing trading, long-term investing
- **Custom signal thresholds:** User-defined boundaries for signal classification
- **Indicator toggles:** Enable/disable specific indicators
- **Timeframe flexibility:** Support for multiple timeframes (intraday, daily, weekly)

11.4 Phase 5: Advanced Features

- **Divergence detection:** Identify price-indicator divergences automatically
- **Multi-timeframe analysis:** Combine signals from multiple timeframes

- **Comparative analysis:** Compare stock vs. sector vs. market indices
- **Historical signal tracking:** Track signal changes and performance over time
- **Backtesting capability:** Test strategy performance on historical data
- **Alert system:** Notifications when signals change or conditions met

11.5 Phase 6: Integration Features

- Fundamental data overlay (P/E ratios, earnings dates)
- News sentiment integration
- Sector rotation analysis
- Market regime detection (bull/bear/sideways)
- Correlation analysis between holdings

Appendix A: Glossary

Moving Average: Average price over a specified time period, used to smooth price data and identify trends

Momentum: Rate of price change, indicating the speed of price movement

Overbought: Condition where price has risen too far too fast, correction likely

Oversold: Condition where price has fallen too far too fast, bounce likely

Crossover: When one indicator line crosses above or below another, often signaling trend change

Volatility: Degree of price variation over time, measure of risk and price movement

Trend: General direction of price movement over time (upward, downward, or sideways)

Signal Line: Moving average of another indicator used for smoothing and generating signals

Divergence: When price and indicator move in opposite directions, often precedes reversals

Support: Price level where buying interest historically prevents further decline

Resistance: Price level where selling interest historically prevents further advance

Breakout: Price movement through established support or resistance level

Consolidation: Period of sideways price movement, often before a breakout

Volume: Number of shares traded, confirms strength of price movements

Band Width: Distance between upper and lower Bollinger Bands, indicates volatility

Appendix B: Technical References

Indicator Calculation Standards

- **Simple Moving Average (SMA):** Sum of closing prices / Number of periods
- **Exponential Moving Average (EMA):** Weighted average giving more weight to recent prices. Multiplier = $2/(periods + 1)$
- **RSI:** $RSI = 100 - (100 / (1 + RS))$, where $RS = \text{Average Gain} / \text{Average Loss over period}$
- **MACD:** MACD Line = $EMA(12) - EMA(26)$; Signal Line = $EMA(9)$ of MACD Line
- **Stochastic:** $\%K = (\text{Current Close} - \text{Lowest Low}) / (\text{Highest High} - \text{Lowest Low}) \times 100$
- **Bollinger Bands:** Upper = $SMA(20) + 2\sigma$; Lower = $SMA(20) - 2\sigma$

Industry Resources

- Technical Analysis Library: Standard technical indicators calculation methods
- Murphy, John J. "Technical Analysis of the Financial Markets" - Industry standard reference
- Investopedia Technical Analysis section - Indicator definitions and interpretations
- StockCharts.com - ChartSchool for technical analysis education

Implementation Notes

- All calculations use closing prices unless otherwise specified
- Volume calculations use actual share volume, not dollar volume
- Moving averages use simple arithmetic mean unless exponential weighting specified
- Standard deviation uses sample standard deviation (n-1 denominator)
- Prices should be adjusted for splits and dividends before calculation

Document Approval

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